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CSE 428

Decoding Anesthetic state from Neural Pixel Data

Exploratory research question advised by Dr. Kevin Schneider at the Golden Lab

Abstract:

General anesthesia (GA) is administered as a sedative in nearly 60,000 surgeries daily in the United States. Yet, there is a very limited understanding about how GA impacts brain activity, leading to induced loss of consciousness and pain sensation. Preliminary work in the Heshmati lab has highlighted key subcortical structures that are engaged during anesthesia, but it remains unclear how activity in these regions and across the brain regulates awareness or pain sensation as anesthesia is induced (“induction”), maintained at a steady state (“maintenance”) and removed (“emergence”), as is done during surgeries.

My work aims to identify the neural circuits that regulate the loss of consciousness and pain sensation during GA by recording action potentials (APs) from mice as they undergo volatile anesthetic isoflurane (ISO).

During AP recordings, small electrodes are inserted into highlighted regions of interest, to capture extracellular action potentials generated by individual neurons and nearby neural populations during the three periods of interest: induction, maintenance, and emergence from isoflurane GA. These action potential recordings are converted to firing rates. I aim to analyze whether this data can be used to decode interval phase from neural activity. In particular, I want to find what regions or timeline are most important to drive accurate predictions.

Through these experiments, I will be able to monitor the effects of isoflurane anesthesia through a temporally-defined electrophysiological lens. Thus, my research aims to further develop our understanding of the brain under GA, by providing novel insight into the neural circuits regulating wakefulness and pain during surgical procedures.

Data:

For each recording, there are 385 channels, or units, along the recording probe. There are several recordings. The data aggregates these recordings such that each data point is a single unit from a recording. For each data point, the metadata notably includes the recording it came from, mouse identification information, and the region information for where the unit was located. Each recording also contains several lists: the firing rate per second, the associated interval phase (baseline, induction, maintenance, or emergence) per second, and the ISO concentration per second. A sample of the recordings is shown:

```
Regions to filter by: ['Amygdala', 'Hypothalamus', 'Hippocampus', 'Thalamus', 'Isocortex']
```

shape: (53, 4)

recording_id	interval_firing_rate	interval_phase	interval_iso
str	list[list[i64]]	list[str]	list[list[f64]]
"NP07_R1"	[[0, 0, … 5], [2, 1, … 0], … [18, 14, … 13]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
"NP07_R1"	[[1, 5, … 0], [3, 4, … 5], … [0, 0, … 0]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
"NP07_R1"	[[6, 13, … 15], [9, 4, … 5], … [6, 4, … 7]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
"NP07_R1"	[[6, 6, … 0], [0, 3, … 2], … [1, 0, … 0]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
"NP07_R1"	[[0, 3, … 0], [2, 1, … 4], … [1, 0, … 0]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
…	…	…	…
"NP06_R1"	[[20, 21, … 26], [31, 32, … 24], … [1, 4, … 8]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
"NP06_R1"	[[18, 10, … 8], [13, 4, … 0], … [0, 0, … 0]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
"NP07_R2"	[[0, 0, … 0], [0, 0, … 0], … [10, 15, … 13]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
"NP07_R2"	[[0, 0, … 7], [3, 6, … 2], … [0, 0, … 0]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]
"NP07_R2"	[[0, 0, … 0], [0, 0, … 0], … [0, 1, … 0]]	["baseline", "baseline", … "emergence"]	[[0.0, 0.0, … 0.0], [0.0, 0.0, … 0.0], … [0.0, 0.0, … 0.0]]

Figure 1: Sample of Data

Methods:

There are two methods that were used over the course of this project: the Logistic Regression Task and the RNN Model Task.

Logistic Regression Task:

A logistic regression model was implemented to establish a baseline for anesthetic phase classification. This baseline approach tests whether static firing rate patterns can distinguish between different anesthetic phases, such as baseline versus maintenance, or induction versus emergence. This approach creates sliding windows across the data such that each data point is processed into an x second long window of firing rate with a single classification phase associated. Two phases are tested to see if they are distinguishable. The data is filtered by region to test the accuracy for that region. The data is also filtered by time (ex. The 0th minute of baseline and 1st minute of maintenance).

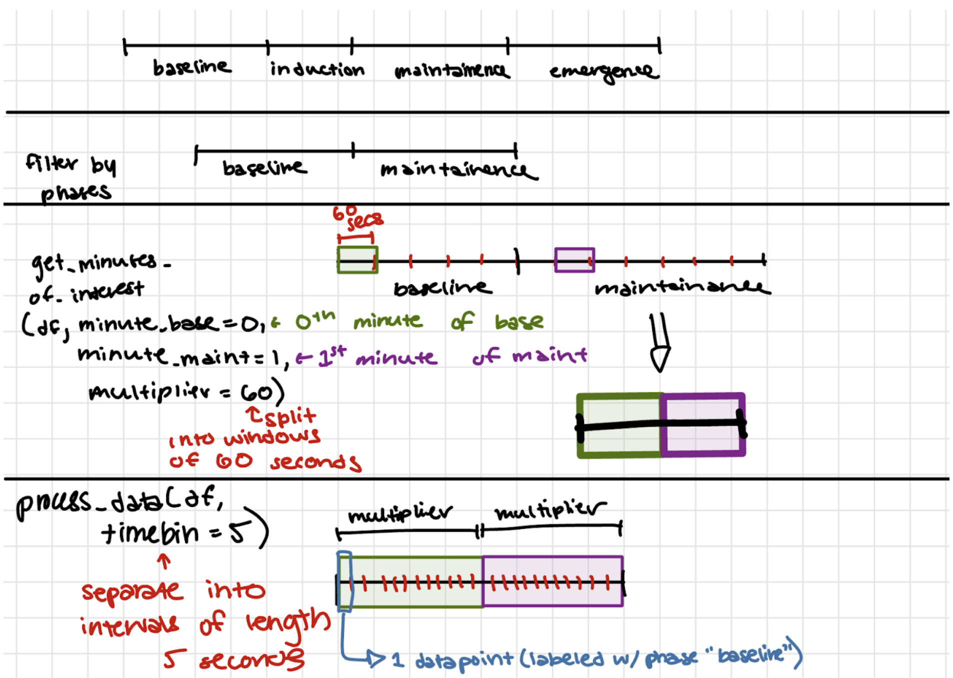

Figure 2: Data processing for the Logistic Regression Task

Then, the data was fed into a logistic regression task. The model was intended to provide a static classification benchmark but does not capture temporal dependencies or sequential patterns in the neural data that may be crucial for understanding anesthetic state transitions. Additionally, this method had too many variables to be able to effectively compare activity in regions or in time. Thus, the next method was created.

RNN Model by Region Task:

A recurrent neural network with attention mechanism was developed to leverage temporal patterns in neural firing rates for anesthetic phase classification. The model architecture consists of LSTM (Long Short-Term Memory) layers that process sequences of firing rates over time, maintaining memory of previous neural states to capture the gradual transitions between anesthetic phases. An attention mechanism was incorporated to automatically identify and focus on the most diagnostically important timepoints, particularly during phase transitions when neural activity patterns change most dramatically.

For this analysis, the data was preprocessed to create standardized 12-minute recordings by extracting specific segments from each phase: the first 3 minutes of baseline, middle 3 minutes of induction, middle 3 minutes of maintenance, and last 3 minutes of emergence. This standardization ensures each phase focuses on stable periods while capturing transition dynamics. Models are created by region to compare the accuracy. This approach ensures balanced representation of each anesthetic phase while capturing stable periods and transition dynamics. The RNN model processes these sequences to make predictions at each second (many-to-many model type), providing high temporal resolution for tracking anesthetic state changes and identifying the neural signatures most predictive of consciousness and pain sensation during general anesthesia.

Results:

Logistic Regression Task:

For each task, several parameters had to be chosen. First, which two phases are being compared. Then, which region the data should be from. Optionally, which window of time from each phase to be filtered for (ex. 0th minute baseline, 1st minute maintenance). Because of the multitude of parameters, I separated the task into two main goals. First, I set the parameters of phase and window and looped through each region to compare accuracy.

```
CURRENT REGION: Hippocampus
CURRENT PHASES: ['baseline', 'maintenance']
Number of each phase: {'baseline': 5880, 'maintenance': 6317}
Accuracy: 0.8553
Confusion Matrix:
[[ 878  297]
 [  56 1209]]

CURRENT REGION: Amygdala
CURRENT PHASES: ['baseline', 'maintenance']
Number of each phase: {'baseline': 1080, 'maintenance': 1920}
Accuracy: 0.6900
Confusion Matrix:
[[ 67 158]
 [ 28 347]]

CURRENT REGION: Isocortex
CURRENT PHASES: ['baseline', 'maintenance']
Number of each phase: {'baseline': 2308, 'maintenance': 4425}
Accuracy: 0.8107
Confusion Matrix:
[[250 206]
 [ 49 842]]
```

Figure 3: Some Logistic Regression Results By Region

In another example, the region and phases were set. Then, the windows of time for the maintenance phase were looped and compared (baseline was set as it should be the same throughout).

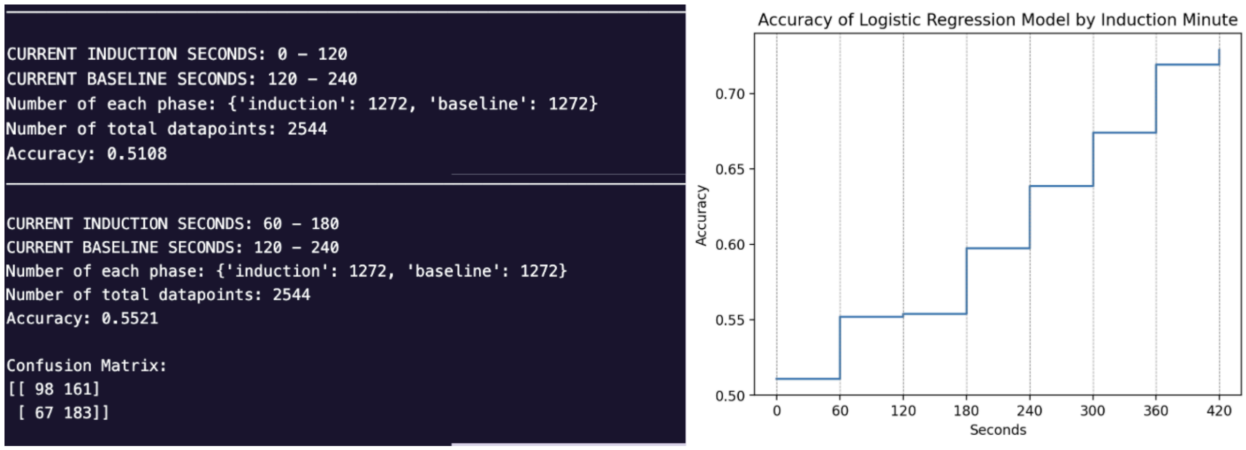

Figure 4: Some Logistic Regression Results by Timebin

RNN Model by Region Task:

Some preliminary results are shown below: For each result, the firing rate activity, attention weights over time, true vs. predicted phase, and attention heat map is shown for each second of the recording. We would predict that the attention weights should be greatest during the transition points of the phases, as these points are the most difficult to detect between and are thus need the most “attention”.

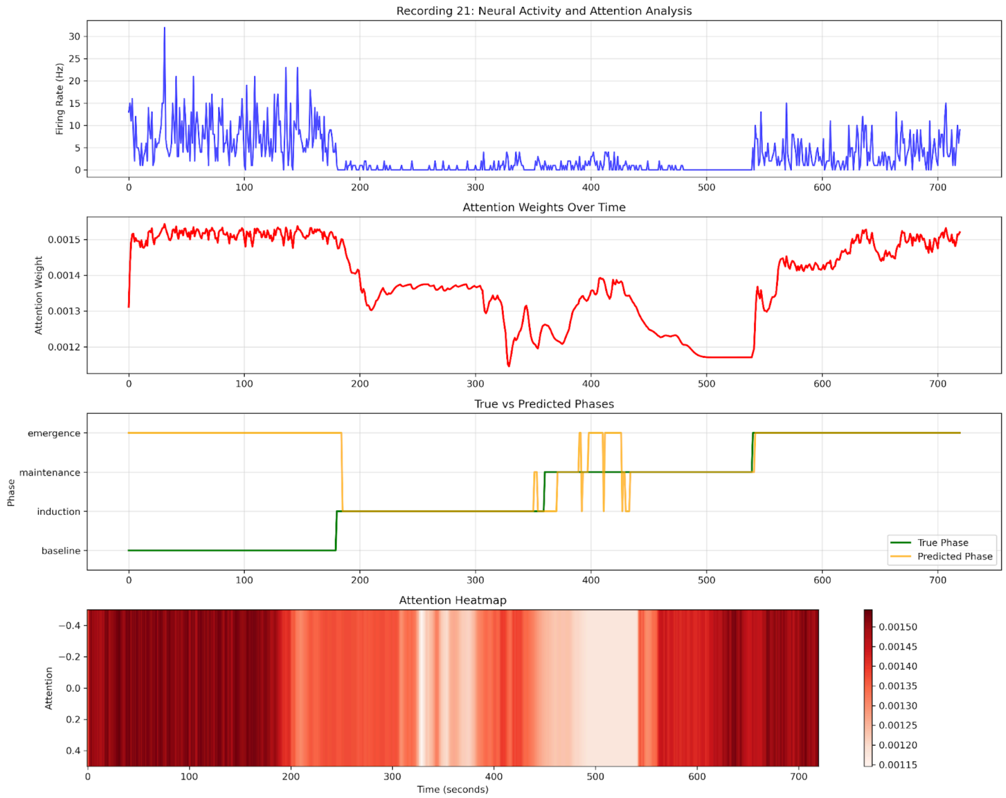

Figure 5: RNN Task for the Hippocampus Region

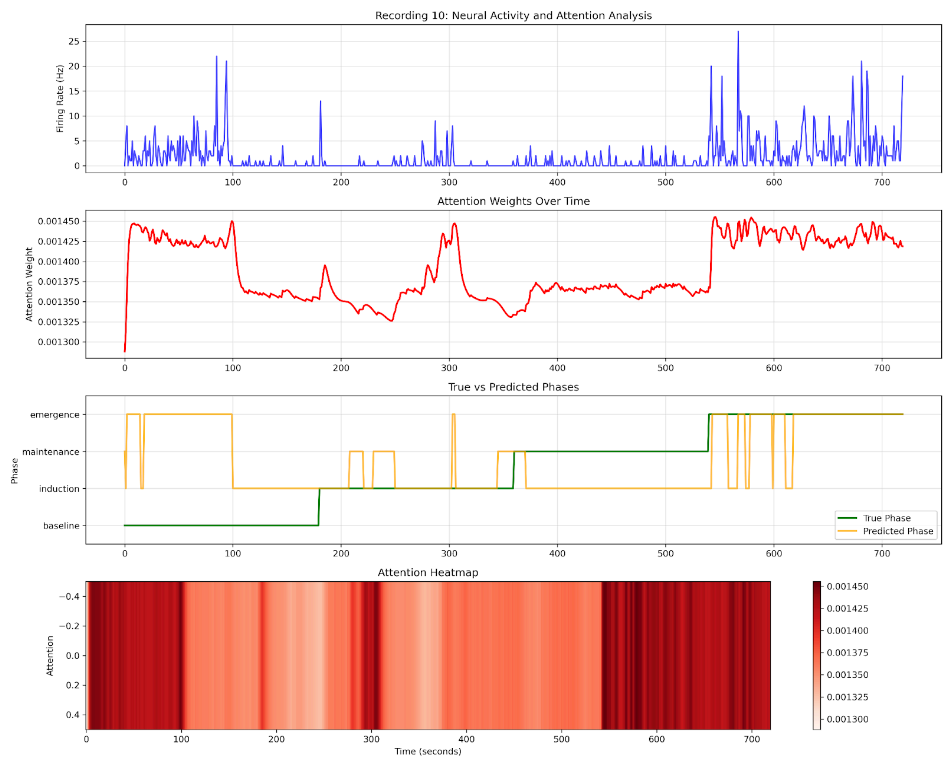

Figure 6: RNN Task for the Isocortex Region

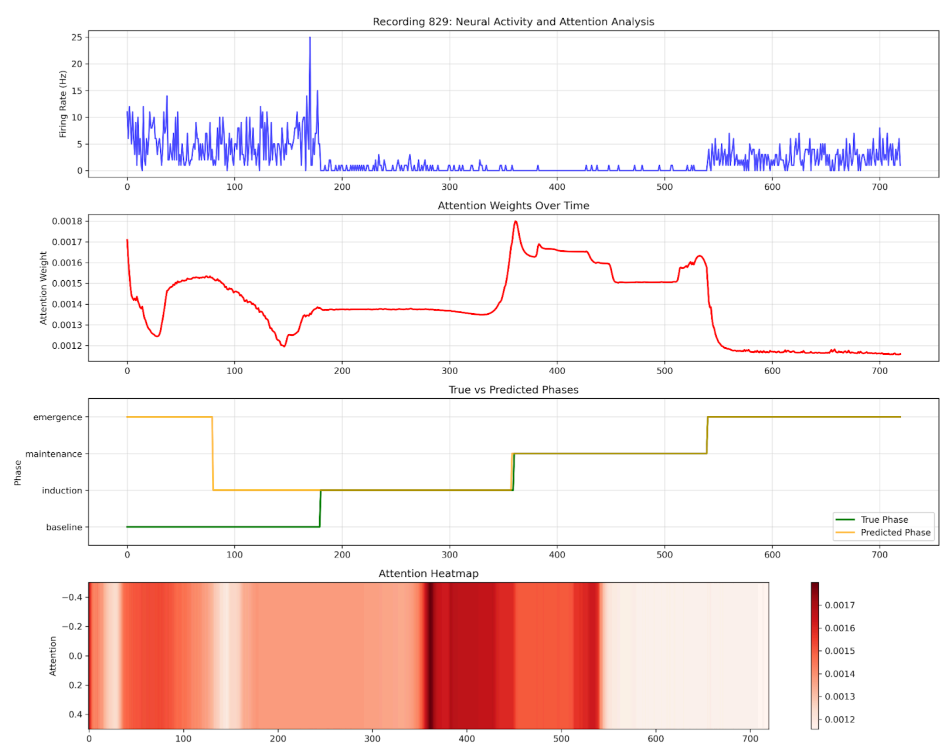

Figure 7: RNN Task for the Thalamus Region

More results for additional regions and representative recordings for each region will be available in the Github folder for this project.

Method	Best Performing Region	Accuracy	Key Finding
Logistic Regression	N/A	Variable	Too many confounding variables
RNN Model	Thalamus	74.42%	Performance correlates with data quantity

Table 1: Results Summary

Conclusions:

While there were many results for the Logistic Regression Task, ultimately no conclusions could be drawn from these tasks as there are too many variables to account for. For example, when comparing the regions it was impossible to know whether a region performed better than another or if that region at that time that was set was more accurate. Additionally, it could not capture the temporal relationships as each data point was a couple seconds long and the relationship between data points in the same unit was lost. Thus, the second task was created to better account for the temporal nature of the data and eliminate the need to filter by windows of time in the data.

Region	Validation Accuracy
Hippocampus	46.91%
Isocortex	38.47%
Thalamus	74.42%
Hypothalamus	47.72%
Amygdala	35.54%

Table 2: Accuracy by Region

It is clear that the thalamus region performed the best from the regions listed with a validation accuracy of 74.42% , compared to the next highest regions hypothalamus (47.72%) and hippocampus (46.91%). However, to further analyze these results, the number of recordings per unit was analyzed.

Region	Number of Unit Recordings
Hippocampus	53
Isocortex	37
Thalamus	883
Hypothalamus	59
Amygdala	16

Table 3: Number of Unit Recordings per Region

The thalamus was 1397% more recordings than the next highest region.

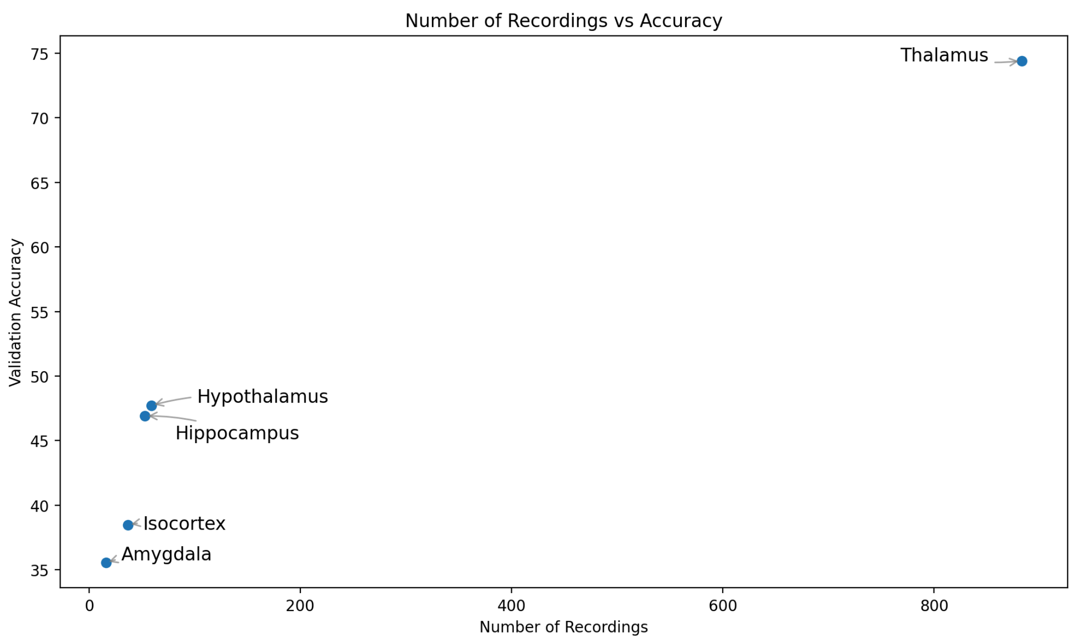

Figure 8: Relationship between Number of Recordings and Accuracy

We can see that there is a positive relationship between number of recordings and accuracy in our current dataset. Thus, it is likely that the thalamus performs much more accurately due to the increase in data rather than fundamental properties of the region itself. Therefore, more data from a variety of regions is required.

Future Directions:

Several key directions will advance this research to better understand neural mechanisms underlying anesthetic state transitions and improve classification accuracy.

Data Collection and Preprocessing Improvements

Future work will focus on expanding the dataset to include more recordings across all brain regions, as current analyses are limited by sample size in certain areas. It is clear that the thalamus, for example, has much more data and thus skews the accuracy comparison across regions. Additionally, to reduce noise in firing rate measurements, temporal binning approaches will be implemented to smooth single-second measurements while preserving important temporal dynamics. Additionally, outlier detection methods will be developed to identify and exclude neural units that exhibit abnormally high activity during induction or maintenance phases, as these may represent artifacts or non-representative cellular responses that could impact classification models.

Model Optimization and Cross-Regional Analysis

Hyperparameter tuning will be conducted to optimize model performance for each brain region. The current approach of training separate RNN models for individual regions will be expanded to enable comprehensive performance comparisons across brain areas, identifying which regions provide the most information for anesthetic state classification.

Multi-Regional Modeling Approaches

To leverage information from multiple brain regions simultaneously, several advanced approaches can be explored. A feature-based method will incorporate brain region identity through one-hot encoding, allowing a single model to process data from all regions while learning region-specific patterns. More sophisticated approaches include embedding techniques that convert categorical region variables into learnable numerical vectors, potentially revealing functional relationships between brain areas and enabling visualization of regional clustering patterns based on anesthetic response profiles.

Advanced Architectural Considerations

Two promising directions for handling multi-regional data include hierarchical modeling and meta-learning approaches. Hierarchical models would first process temporal dynamics within individual regions using specialized RNN architectures, then combine regional outputs through higher-level integration networks. Meta-learning frameworks could extract generalizable temporal features from time series data before incorporating region-specific information at downstream processing stages.

These future directions will enable more comprehensive characterization of neural circuits underlying anesthetic action and potentially inform clinical monitoring approaches for surgical procedures.

Thank you for a great quarter!